# littleangel

Project Report

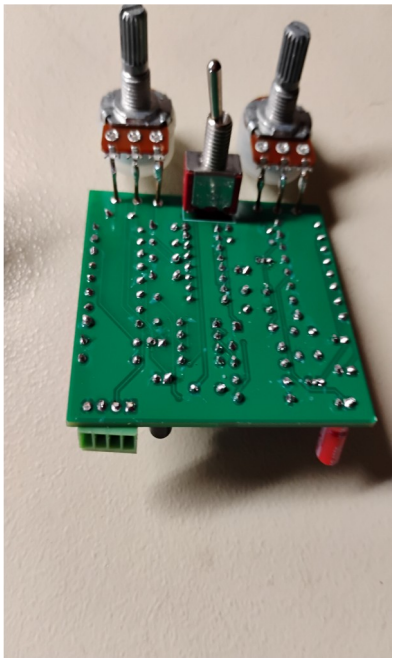
## 1. Readme

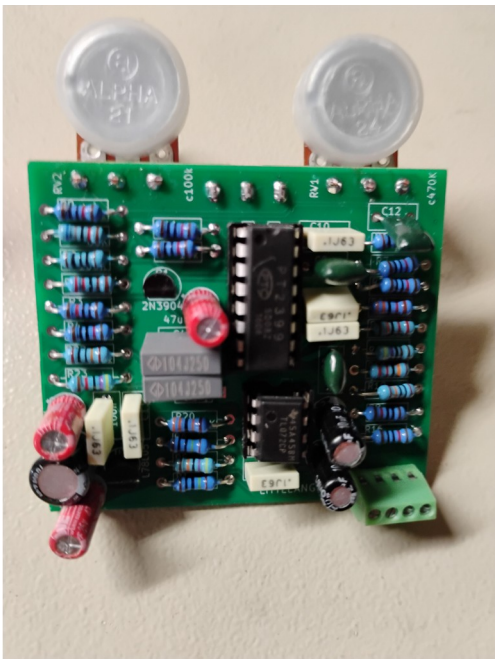
## Little Angel

This is a clone Rick Holt's  Little Angel Chorus.

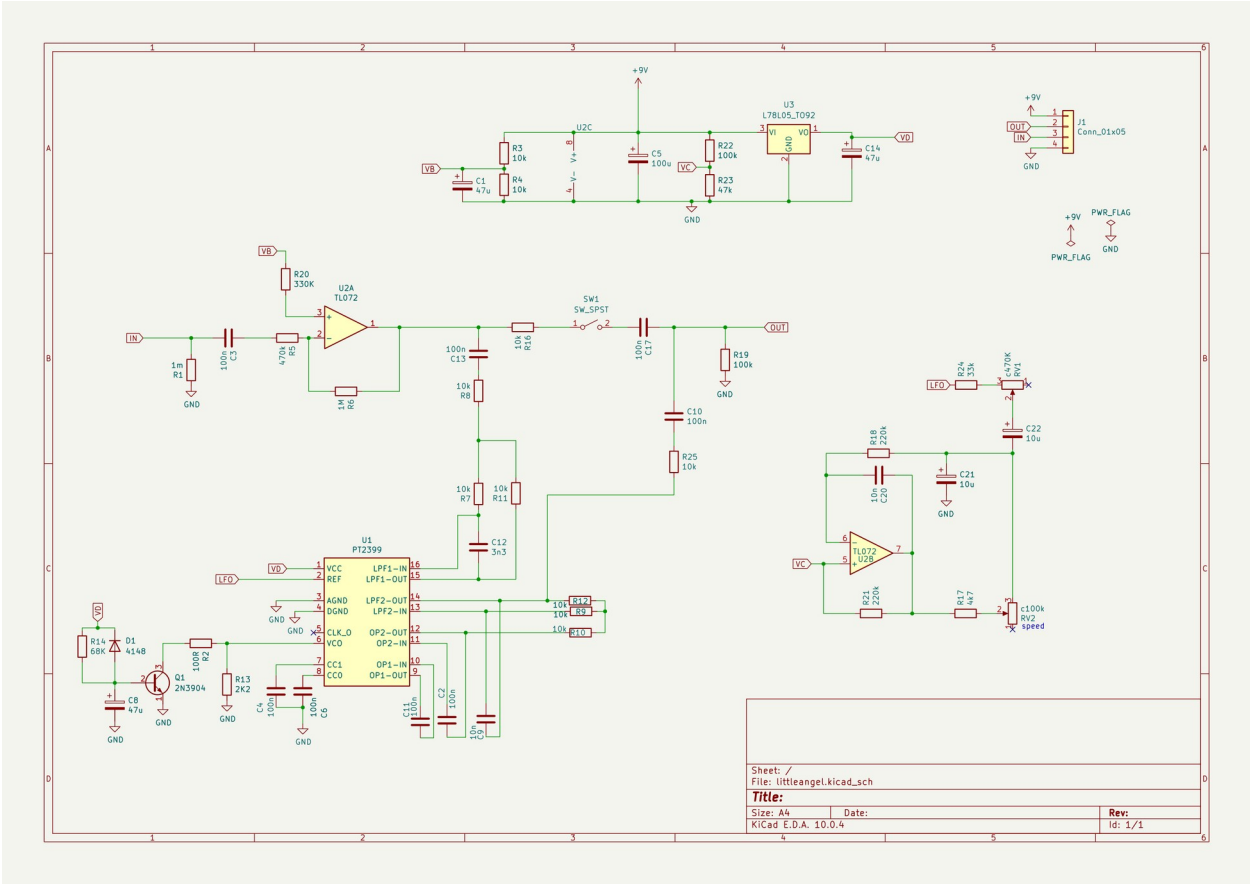
This was based on the a schematic I found on the web for this. THis has the "Vibe Switch Mod" someone did,  which changes between vibrato and chorus modes. Nice pedal.

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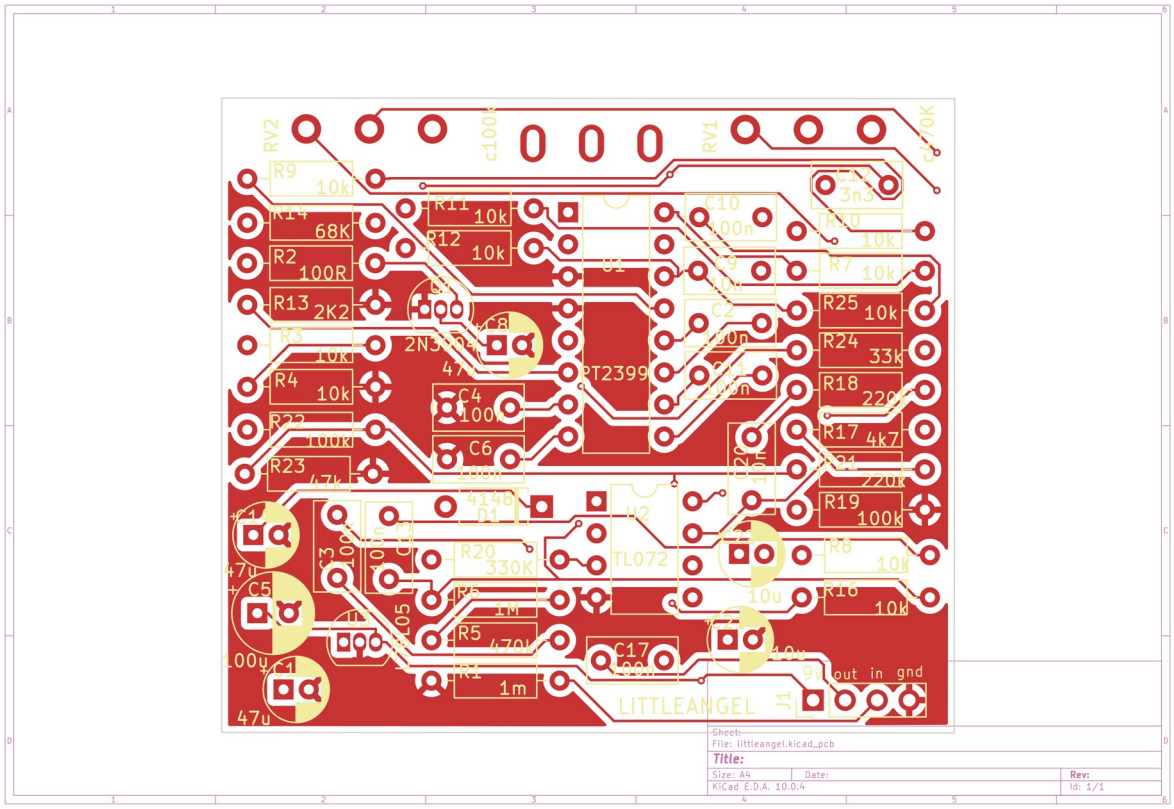




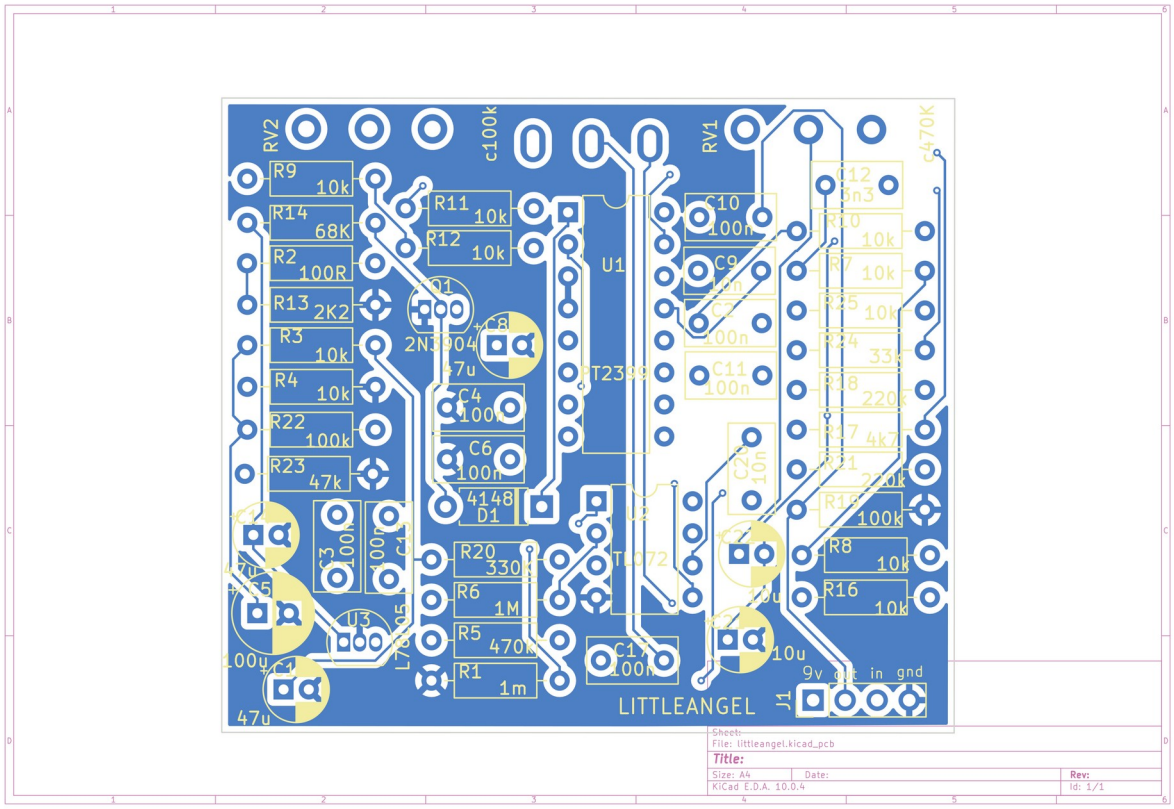
## 2. Schematic


+9V
U3
L78L05_TO92
U2C
R3 10k
R4 10k
C1 47u
C5 100u
R22 100k
R23 47k
C14 47u
VB
VC
VD
GND
J1
Conn_01x05
OUT
IN
PWR_FLAG
R20 330K
U2A
TL072
IN
1m R1
100n C3
470k R5
1M R6
SW1
SW_SPST
100n C13
10k R16
100n C17
R19 100k
OUT
10k R8
C10 100n
R25 10k
10k R7
10k R11
C12 3n3
U1
PT2399
VCC
REF
AGND
DGND
CLK_O
VCO
CC1
CCO
LPF1-IN
LPF1-OUT
LPF2-OUT
LPF2-IN
OP2-OUT
OP2-IN
OP1-IN
OP1-OUT
LFO
R12 10k
R9 10k
R10 10k
R14 68K
D1 4148
Q1 2N3904
100R R2
R13 2K2
C8 47u
C4 100n
C6 100n
C11 100n
C2 100n
10n
R24 33k
470K RV1
C22 10u
R18 220k
10n C20
C21 10u
TL072 U2B
R21 220k
R17 47k
c100k RV2 speed
Sheet: /
File: littleangel.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1

## 3. PCB F.Cu + Silkscreen



## 4. PCB B.CU + Silkscreen



## 5. Bill of Materials

| Reference | Value | Footprint | Qty | DNP |
|---|---|---|---|---|
| **C1,C8,C14** | 47u | Capacitor_THT:CP_Radial_D5.0mm_P2.00mm | 3 | |
| **C2-C4,C6,C10,C11,C13,C17** | 100n | Capacitor_THT:C_Rect_L7.0mm_W3.5mm_P5.00mm | 8 | |
| **C5** | 100u | Capacitor_THT:CP_Radial_D6.3mm_P2.50mm | 1 | |
| **C9,C20** | 10n | Capacitor_THT:C_Rect_L7.0mm_W3.5mm_P5.00mm | 2 | |
| **C12** | 3n3 | Capacitor_THT:C_Rect_L7.0mm_W3.5mm_P5.00mm | 1 | |
| **C21,C22** | 10u | Capacitor_THT:CP_Radial_D5.0mm_P2.00mm | 2 | |
| **D1** | 4148 | Diode_THT:D_A-405_P7.62mm_Horizontal | 1 | |
| **J1** | Conn_01x05 | Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical | 1 | |
| **Q1** | 2N3904 | Package_TO_SOT_THT:TO-92_Inline | 1 | |
| **R1** | 1m | Resistor_THT:R_Axial_DIN0207_L6.3mm_D | 1 | |

| | | | | |
|---|---|---|---|---|
| | | 2.5mm_P10.16mm_Horizontal | | |
| **R2** | 100R | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **R3,R4,R7-R12,R16,R25** | 10k | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 10 | |
| **R5** | 470k | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **R6** | 1M | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **R13** | 2K2 | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **R14** | 68K | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **R17** | 4k7 | Resistor_THT:R_Axial_DIN0207_L6.3mm_D | 1 | |

| | | | | |
|---|---|---|---|---|
| | | 2.5mm_P10.16mm_Horizontal | | |
| **R18,R21** | 220k | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 2 | |
| **R19,R22** | 100k | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 2 | |
| **R20** | 330K | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **R23** | 47k | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **R24** | 33k | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal | 1 | |
| **RV1** | c470K | pedals:Pot_Underside | 1 | |
| **RV2** | c100k | pedals:Pot_Underside | 1 | |
| **SW1** | SW_SPST | pedals:SWITCH-M1-SPDT | 1 | |
| **U1** | PT2399 | Package_DIP:DIP-16_W7.62mm | 1 | |

| | | | | |
|---|---|---|---|---|
| **U2** | TL072 | Package_DIP:DIP-8_W7.62mm | 1 | |
| **U3** | L78L05_TO92 | Package_TO_SOT_THT:TO-92_Inline | 1 | |

Total line items: 28